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SHORT-PAPER

VR4VRT: Virtual Reality for Virtual Rowing Training

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VR4VRT: Virtual Reality for Virtual Rowing Training

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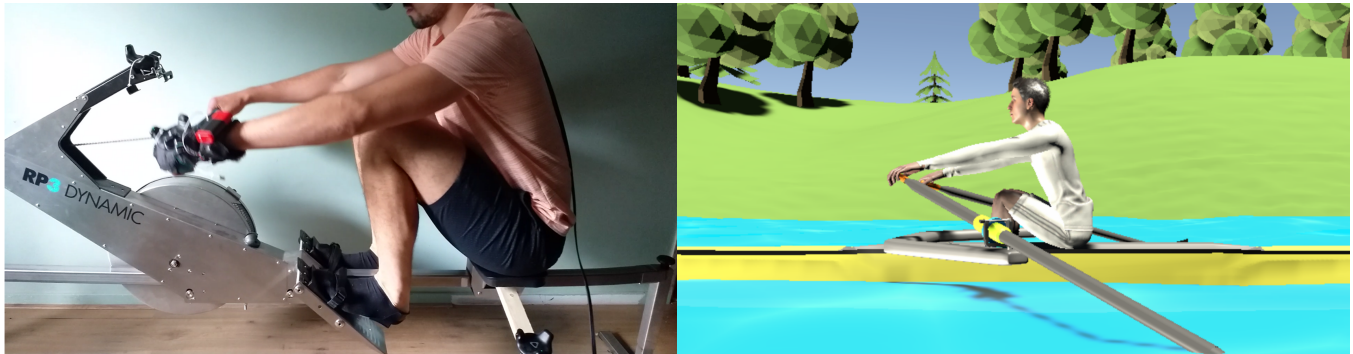


Figure 1: VR4VRT: Virtual Reality for Virtual Rowing Training. The user wears a head-mounted display and a motion tracker on a glove with built-in haptic feedback, and sits on an ergometer augmented with two motion trackers (seat and moving flywheel). This results in the virtual side view (e.g. for a coach or replays), whereas the user sees a first person view [3].

ABSTRACT

Learning how to properly row is an intensive and injury prone process. Ergometers (i.e., rowing machines) are used as a preparatory step to train the basic steps of rowing, as well as to further improve technique and maintain stamina. In this paper we present three design iterations resulting in a new interactive system that utilizes the commercially available dynamic ergometer (RP3). Our proposed software-hardware combination uses the RP3 with the HTC Vive platform extended with three location trackers, from which the software is available on request. We will showcase how this facilitates a range of opportunities.

CCS CONCEPTS

• **Human-centered computing** → *Systems and tools for interaction design*; Haptic devices; **Virtual reality**; • **Software and its engineering** → **Virtual worlds training simulations**.

KEYWORDS

Rowing, Virtual Reality, Sports, Training, Ergometer, HMD

* Authors contributed equally, we build on the 2nd & 3rd authors' BSc theses [2, 30].

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1 INTRODUCTION

Unfortunately, injuries are common in rowing: 32% to 51% of rowers experience strain in the lumbar spine [31] due to intense training schedules and incorrect technique of beginning rowers [19]. Coaching is needed for beginners to learn the right technique. However, a coach can give each rower in a group only a fraction of their time, and can even be biased towards athletes they know well [11]. Furthermore, during training there may be several complete beginners alongside more experienced rowers, which complicates coaching.

Training rowing on water is not always possible, because of weather, temperature, and teammate availability. Therefore, rowing indoors on an ergometer is an important part of the sport as it presents more flexibility in scheduling training sessions.

Most ergometers display numbers and graphs. These are quite useful if one knows how to read them, but a beginner does not, and it is impossible for a coach to analyze everyone's data in real-time. Also, even for experienced rowers it is challenging to continuously be aware of and interpret all variables to optimize the rowing stroke. Thus, there is a need and room for improvement to make this information useful to beginning and experienced rowers.

Using interactive technology, we want to make it easier for beginners to optimize the rowing stroke and to solve the earlier mentioned coaching difficulties. Addressing this with an off-the-shelf

VR solution simultaneously makes it possible to enable more research and to widely distribute the system in working form.

When applying VR to sport, the environment can be controlled in specific and reproducible ways [13] to assess performance, provide continuous feedback, and practice specific skills. An impressive body of academic work to build upon already exist on structuring multi-modal feedback systems for rowing technique [1, 9, 20–22, 25, 26, 32]. Feedback during and after training is essential for athletes to know what good and bad technique feels like [11], and with VR this feedback can be given via extremely immersive visual channels, on top of auditory and wearable vibrotactile haptic channels.

This paper offers **two main contributions** to the CHI PLAY community. **One**, the system combines current ways of autonomous feedback for training to improve a beginner's rowing technique, and the explorations we present show its adaptability. Thereby, the VR4VRT as an artefact embeds the value of how it can help training and might decrease the risk of injuries, while at the same time providing unsupervised learning of rowing technique. In turn, our work can inspire future investigations into rowing training. **Two**, it is an access point to a dynamic software platform that can be used for a variety of research goals into rowing, in the real-world with longer term testing due to its low-cost and easy setup¹.

2 RELATED WORK

Existing work addresses the same context through sonification, haptic technology, visual feedback on displays or advanced immersive CAVE-systems, and combinations thereof. We present a short overview of systems we build on.

Sofirow is for in-boat use with accompanying measurement device Accrow and provides audio feedback [12, 25, 26]. It maps the acceleration of the boat onto tone pitch. For the rowers, wearing earpieces, this can increase perception of details in the rowing movement, and studies show a significant positive effect on velocity of the boat [25].

Building on this, Effenberg et al. provide ‘*continuous 4-dimensional movement acoustics on two kinematic and two dynamic movement parameters*’ [9, p5]: stroke length, seat position, footrest force, and handle force. Their study shows positive effects of sonification regarding measured footrest force and seat position.

Fothergill et al. [10] describe a visual feedback system that provided graphs based on 3D motion capture, force, and power measurements, related to four different parameters: height of handle, footrest force, handle force, and horizontal distance between seat and handle. Their small-scale study ($n = 5$) hints towards training of consistent good technique.

Anderson et al. project a side-view video stream in front of the user [1]. This is combined in overlay with a pre-recorded expert execution as visual feedback during training. Their study results ($n = 16$) show a significant power increase for participants that received this concurrent feedback. Note that they use a set of pre-defined stroke rates for the accompanying expert videos.

Two systems combine motion sensing with a form of immersive feedback. The *SPRINT Rowing Training System* aims to improve

technique, energy management and team coordination of rowers [21, 22]. It provides augmented visual, auditory, and haptic (vibrotactile) feedback to optimize technique, and trains sculling (two oars) and sweep rowing (one oar). It can be configured into an immersive CAVE setup where the rower's motion is measured with Vicon motion capture. In the *M³* (Multi-Modal Motion synthesis) simulator the rower interacts using haptic oar movement and simultaneously perceives results visually and acoustically [20, 32]. This system uses force feedback rather than vibrotactile feedback, provides auditory cues through a ring of loudspeakers, and fully surrounds the user, allowing the user to look to the side to observe their oars at the place they would be in the real environment.

There are also several inspirational commercial products. *Quiske* offers an app and a wireless pod placed on the ergometer to measure five metrics including rhythm, legs speed, and (seat) stops. *BioRowTech* uses a set of dedicated sensors, including a measuring wire attached to clothes, handle, and seat, to provide visual feedback on three often made errors including grabbing of the arms, moving the back too early, and excessive movement of the back at the finish [17]. *Holofit - Holodia* [14] is a VR application for rowing, tracking seems to be based on the headset, and is mainly entertainment-based (e.g. a boat going in the wrong direction [24]).

In summary, we see effective interventions regarding haptics, sonification, and visual feedback. Some systems offer a simple and low-cost solution; some focus on including multiple modalities. We see an opportunity for research purposes to offer both: adaptable, modular combinations of modalities, based on low-cost technology combined with a common high-end ergometer. In turn such a platform can facilitate flexibility for longer term testing at rowing clubs or at home (e.g. skill transfer beyond lab visits, where CAVE-like systems might be difficult to install at various locations) and to investigate technology-specific interventions (e.g. new variations of on-location feedback specific for VR). We report our investigation into this with the iterative design of VR4VRT, to this end we first discuss aspects of rowing technique that the system should incorporate in measurement and feedback.

3 ROWING TECHNIQUES TO EMBED

For our view on rowing techniques we build on own experiences (including a competitive rower), interviews with coaches, coaching manuals from the local rowing club [29] and the national rowing association [18], and videos from professional rowers [6, 8].

The rowing cycle is commonly divided into four phases: 1) Catch, 2) Drive, 3) Finish, and 4) Recovery, see Figure 2 (cf. [26]: front reversal, drive, back reversal, recovery).

The *catch* is the starting position, from which the *drive* starts with a push of the legs, followed by a swing of the back, and *finish* by the arms. This order is important but they should not be performed as completely distinct steps, see Figure 2. The rowing club's manual [29] and interviewed rowing coaches mention a very common misunderstanding for beginners: that the handle is supposed to be pulled with the arms, however this is detrimental to stroke efficiency and quickly exhausts the relatively weak arm muscles.

After the finish, the *recovery* starts in which the rower moves towards the next stroke and executes the same movements in reversed order. While being a relaxed slower movement, it is important to

¹The code will be shared on request without any costs, for the foreseeable time, under the well-known CC 4.0 CC-BY-SA terms: specifically attributing this paper. Requests can be made to the two main authors over email, in which one has to detail the planned use and to commit to these terms of use.

carefully execute the technique to smoothly prepare for the catch. Now the arms and back do have overlap, and there is also overlap between back and legs. An important part to get right here, also for injury prevention, is finishing the incline of the back about half-way through recovery. This way the rower only has to bend their legs further and has a strong core and lower back at the catch.

Finally, the *trajectory* of the hands is important, especially during on-water rowing (cf. [23]). It is necessary for efficiently moving the oar in and out of the water, and for achieving stability. When rowing indoors, trajectory is still important, besides its importance for transferring skills to the water, as the height of the handle is crucial at the catch and should be pulled in a straight line during the entire drive. This is important for power efficiency on an indoor rowing machine as then the handle is at the same height as the chain attached to the machine.

3.1 Interviews and Expert Input

We also identified the most common mistakes and how to overcome these as indicated in [18]. We had an orientating discussion about technique with a familiar coach, then our first interview with 2 coaches, and half a year later an interview with 4 coaches (in total 5 coaches helped with 2x3 years, 1x6 years, and 2x8 years of experience). One of the coaches nicely summarized four main groups of errors: *Catch*, legs and core not tightened; *Drive*, not a straight handle trajectory; *Recovery*, wrong order of execution; not achieving the ideal stroke length, *angle of the back* should be sufficiently inclined, arms should be stretched, and catch should be deployed when the shins are vertical. Regarding human *feedback* with which to improve, different modalities were mentioned: executing the movement as an example, putting people in a stance, letting them freeze their posture, guiding their movement, using verbal feedback, or audio cues for timing². Most coaches agreed that the ‘most effective feedback’ cannot really exist as feedback methods needs to be alternated. Additionally, there are very different types of rowers which all learn most effectively in their own way.

3.2 Four Elements for Training & Feedback

All in all, the above made us focus on four elements that our system will provide feedback on. *Overlap*: the first target element for feedback is the overlap, related to execution order in the recovery and to some extent the timing during the recovery. *Velocity of hands*: for the chosen range of row stroke rate and boat-type simulation, the rower should maintain approximately the ‘standard’ 1:2 ratio of hand velocity between drive and recovery [5, 15, 16]. *Handle trajectory*: this should be straight in the drive, not ending too high, and with an appropriate curve in the recovery. This also helps to correct errors such as ‘missing the catch’, ‘not a straight line during the drive’, ‘finishing too high or too low’, and ‘not having the ideal stroke length’. *Angle of the back*: we will indicate when the rower should move their back forward or backward and to what extent.

4 VR4VRT

VR4VRT is an interactive ergometer trainer based on off-the-shelf components: an RP3 dynamic rowing machine in combination with

²Interesting is a recent radio broadcast including in-boat audio recordings and interviews of (professional) rowers [7]

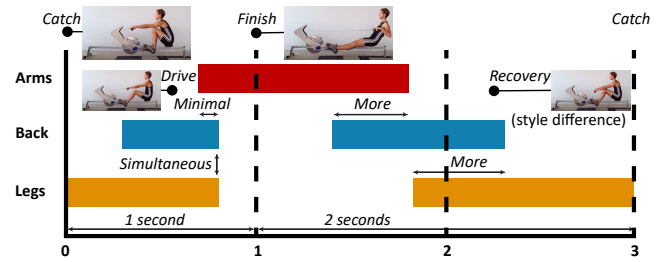


Figure 2: Timing [8] of the movement of legs, arms, and back, and posture [6]. There are different styles; timing is based on a video of Australian two-time World Champion, three-time Olympic gold medalist Kim Crow [8]. Her style is quite representative of how rowing is taught to beginners.

the HTC Vive (2016) and accompanying motion tracking devices. The platform is built to work in a modular fashion where types of feedback can be turned on or off and modes of feedback can be switched. We used Unity with SteamVR, integrating several existing assets and plugins for models and behavior of character, boat, environment, and feedback. By building on top of such highly accessible and available system components we aim to make the resulting system easy to obtain and deploy in the context of other research projects (e.g. beyond the lab). In this section we explain the key aspects.

4.1 Trackers and Placement

Our first basic version used a standard ergometer and only the Vive controllers hooked up to the handle of the ergometer, making it possible to respond mainly to the height of the handle. Although we do have access to high-end mocap platforms in our labs, such as OptiTrack and XSense, these lack in easy calibration, comfort (i.e. a suit), cost, and maintenance as well as required expertise. Instead, using the off-the-shelf VR setup, VR4VRT targets to also test at rowing clubs with a reliably accurate yet easy to set up platform.

We looked at a trade-off between the number of trackers and their accuracy of tracking. Since joints in the middle of a chain can be estimated using inverse kinematics, the hands and feet of users are one of the most important to track right. We assume that the user always has their feet in the ergometer footrests and their hands holding the handle. This means that the distances between hands and between feet should stay the same. Tracking the ergometer handle and footrests are therefore sufficient to get an accurate enough representation of limb posture.

In the third iteration we switched from tracking the handle to attaching it onto a glove, see Figure 1, as the handle vibrations caused inaccurate motion tracking. To prevent occlusion, we placed the tracker on top of the flywheel. We used a similar principle on the other side of the limbs: by tracking points between the shoulders with GoPro belt, for the arms, and underneath the lower back (i.e. the seat), for the legs. Instead, in the third iteration we also assume a straight head and neck posture, thus with the head position and orientation (i.e. the head-mounted display) we can estimate the center of the shoulders position, and the so important back angle. This means no belt and one less tracker is needed, at the cost of

the ability to compare VR to non-VR with automated tracking (i.e. fair comparison in automatic measurement with no HMD). These adjustments of the third iteration resulted in a noticeable improvement in stability.

Furthermore, due to COVID-19 we had to switch from lab to home use, also relevant for smaller research or rowing rooms. We tried out various options to prevent occlusion and improve tracking. In the end we placed the trackers high up to the non-wall side, and added a small wooden plank under the seat on the non-wall side, for which we used cable ties and double sided tape for the placement, see Figure 1. The user must always pair the trackers by turning the trackers on in the right order: Flywheel, Hand, Seat. For each new room setup a short ‘calibration’ is done. During calibration of the room the headset is to be placed on the seat with the seat at an indicated place (e.g. with tape).

4.2 Modular Extensions

In order to show the versatility of VR4VRT we integrated and addressed various suggestions from the related work. We addressed **overlap** in our second iteration feedback system, which we still have to analyze using more appropriate statistical analysis and filtering [30]. Inspired by [4], we used a simplified real-time depiction to indicate timing and order of movements *during the recovery* of rowing, similar to the right side of Figure 2, but we overlaid the performance of the last stroke over this goal timing.

In the third iteration, we addressed the **angle of the back**; similar to [1] we also showed an overlay of an ‘expert’s’ execution. We made use of a full-body depiction with a red-colored expert avatar made transparent to various degrees (i.e. bandwidth feedback, that is the further away from the ideal situation the more feedback one got [27]). We also made a simplified version where we just showed a capsule indicating the back. Unlike [1], we can also measure the stroke rate from the previous strokes, which in the future also allows to automatically adapt the visualization.

Additionally, we addressed **velocity** and **trajectory** with multiple designs for visual, audio, and haptic feedback. We were able to visualize the velocity and trajectory at the handle of the oar, to the side of the user, as a head-up-display relatively placed overlay, and as an in-front placed side view (cf. [1, 28], instead we provide new locations and types of feedback). As can be seen in our accompanying video [3], the *visualization of trajectory* as a drawn curve over a target and the timing/velocity in the shape of a ball moving over this curve was inspired by [28]. Here we also used bandwidth feedback [27], for instance, increasing the loudness to a maximum if the handle touches the chin; or making a curve fully transparent when a movement is done properly. This way users might have only focused on errors when necessary.

Building on Unity and FMOD sound effects we tried out *spatial sound* (e.g. whether the target is behind or in front of the user’s *velocity*) with varying pitch for *trajectory*. Based on suggestions in a Unity forum³ we implemented a low pass filter and reverb to emphasize the sound coming from the rear. Similar to Sofrirow with Accrow, we used separate notes, in our case to the height of the handle indicating the trajectory. We were able to vary this between absolute (e.g. high handle position gives high pitch) or relative (i.e.

deviation from target position in pitch). We also implemented a version where we played the target pitch in the left ear and the current pitch in the right ear.

To test various forms of *haptic feedback for trajectory* (e.g. steering towards right height by vibrating under or above the palm of the hand) we used two vibration motors and additional batteries connected to a ESP32 interacting wireless with the Unity server detecting the performance, see [2] for details. We varied whether people should be ‘pushed’ or ‘dragged’ into the right positions as this was not clear from related work. We managed to make the vibrations noticeable, but the varying intensities were not. We also added a few additional extensions. Currently we have two environments, a virtual fitness room or an on-water simulation.

5 DISCUSSION

With VR4VRT we showed the applicability of a low-cost motion capture system to investigate rowing techniques and new ways of training. We managed to provide various ways of multi-modal autonomous concurrent feedback and offer an on-request available software platform accompanied with instructions for calibration and ideas for future investigations.

Our next focus is on four follow-up developments. One, we will add integration of power curves directly from the RP3. Two, we will add more standardized calibrations steps in the software platform (i.e. easily adjusting distance specific markers). Three, we add user profiles, which will allow a bigger variety than our current two body proportions standard, this should also be linked to an optimal trajectory size. Four, we will investigate usability aspects preparing for longer term user testing, such as varying HMD covers to deal best with sweating and larger number of users.

Our next research steps include analyzing our second iteration user test and the improvements based on our third iteration qualitative user encounters in appropriate detail to generate broader design lessons and applicability and limitation of our approach. We will then investigate two directions. One, we will work towards more entertainment-oriented avenues adding additional game-elements such as competition, perhaps linked with training schedules. Two, building on this work and others regarding running, we are starting a cross-discipline multiple PhD university funded research theme: Sports Data and Interaction. Based on data science, statistics, and bio-mechanic modeling we will investigate anomalies in execution in more detail to provide appropriate concurrent (during) and terminal (after) feedback for coach and athlete.

With VR4VRT and the accompanying paper we provide an additional, to be tested, method for unsupervised learning of rowing technique, hopefully lowering the chance of injuries, heightening performance, improving experience, and resolving coaching problems in groups with varying skill levels.

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³<https://forum.unity.com/threads/3d-audio-positioning-front-rear.44697/>

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